Name: WAQAR RIASAT ALI Section: E Roll num: 2023F - BSE - 221

LAB NO 14 (OPEN - ENDED)

DATA STRUCTURES AND ALGORITHMS

OBJECTIVE : Create a program that uses the Queue data structure to simulate a grocery store billing counter. The system should add customers to the queue, process them in order, and display the queue's status

Input 01:

```java
package lab_14;

import java.util.LinkedList;
import java.util.Queue;
import java.util.Scanner;
public class LAB_14 {

    public static void main(String[] args) {
        Queue<String> queue = new LinkedList<>();
        Scanner scanner = new Scanner(System.in);

        while (true) {
            System.out.println("\n--- Grocery Store Billing System ---");
            System.out.println("1. Add a Customer to the Queue");
            System.out.println("2. Serve a Customer");
            System.out.println("3. View the Next Customer");
            System.out.println("4. Display All Customers");
            System.out.println("5. Exit");
            System.out.print("Enter your choice: ");
            int choice = scanner.nextInt();
            scanner.nextLine(); // Clear the input buffer

            switch (choice) {
```

```java
                case 1:
                    System.out.print("Enter customer name: ");
                    String name = scanner.nextLine();
                    queue.add(name);
                    System.out.println(name + " added to the queue.");
                    break;

                case 2:
                    if (queue.isEmpty()) {
                        System.out.println("No customers in the queue to serve.");
                    } else {
                        System.out.println("Serving " + queue.poll());
                    }
                    break;

                case 3:
                    if (queue.isEmpty()) {
                        System.out.println("No customers in the queue.");
                    } else {
                        System.out.println("Next customer: " + queue.peek());
                    }
                    break;

                case 4:
                    if (queue.isEmpty()) {
                        System.out.println("No customers in the queue.");
                    } else {
                        System.out.println("Current queue: " + queue);
                    }
                    break;

                case 5:
                    System.out.println("Exiting the system. Goodbye!");
                    scanner.close();
                    return;

                default:
                    System.out.println("Invalid choice. Please try again.");
            }
        }
    }
}
```

Output 01:

```
--- Grocery Store Billing System ---
1. Add a Customer to the Queue
2. Serve a Customer
3. View the Next Customer
4. Display All Customers
5. Exit
Enter your choice: 1
Enter customer name: WAQAR
WAQAR added to the queue.

--- Grocery Store Billing System ---
1. Add a Customer to the Queue
2. Serve a Customer
3. View the Next Customer
4. Display All Customers
5. Exit
Enter your choice: 2
Serving WAQAR

--- Grocery Store Billing System ---
1. Add a Customer to the Queue
2. Serve a Customer
3. View the Next Customer
4. Display All Customers
5. Exit
Enter your choice: 5
Exiting the system. Goodbye!
BUILD SUCCESSFUL (total time: 2 minutes 5 seconds)
```